LoRa Peer-to-Peer Communication Lab

Objectives

- Familiarize with LoRa technology.
- Develop a LoRa-based peer-to-peer solution using Arduino and Cytron LoRa-RFM Shield.

Equipment

- Computer / Laptop
- Maker Uno
- Cytron LoRa-RFM Shield

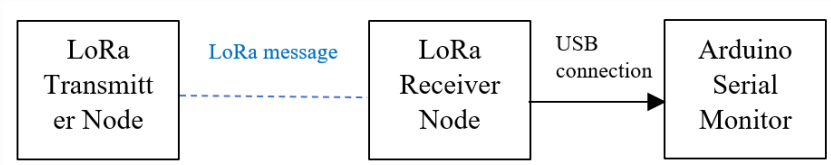
Introduction

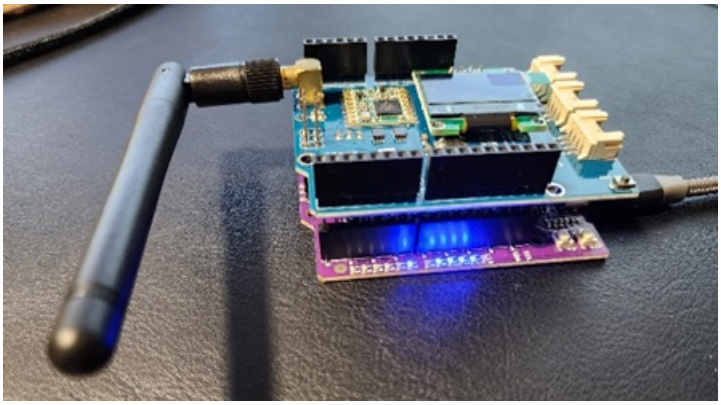
LoRa (Long Range) is an RF modulation technique designed for low-powered wireless communication systems, offering large transmission ranges (up to 16 kilometers in rural areas) due to its use of lower frequencies compared to standard wireless methods like Wi-Fi. It's ideal for battery-powered IoT devices in applications requiring communication over long distances. LoRa networks uses a peer-to-peer topology that can be scalable and operate in the license-free ISM frequency band, making it a cost-effective solution for IoT applications. This lab aims to provide hands-on experience with LoRa devices in a peer-to-peer (P2P) configuration, highlighting its potential in IoT applications.

Exercise 1: LoRa Peer-to-Peer Communication

In this exercise, we will use Maker UNO and Cytron LoRa-RFM Shield to build a LoRa-based end node for data exchange in a peer-to-peer setup. We'll create a transmitter node broadcasting LoRa payloads and a receiver node to display this data on a serial monitor.

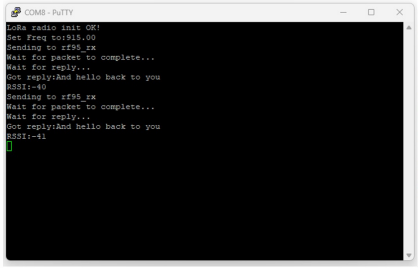
Ensure that the Maker UNO is connected to the Cytron LoRa-RFM Shield, as shown below.

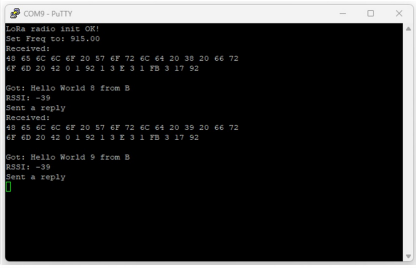




Setup

1. **Arduino IDE Setup**
   - Download and install Arduino IDE (preferably version 1.8x).
   - Install the CH341 Serial Driver for Maker UNO.
   - Install the the open-source library RadioHead by Paul Stoffregen. This will allow us to send and receive data using LoRa-based radios.
   - In Arduino IDE, select `Tools > Board > Arduino Uno`.
   - Connect the Arduino to the USB port and select the correct Serial Port under `Tools > Port`.
2. **Programming LoRa Communication**
   - Use provided examples (`rf95_rx.ino` for Receiver and `rf95_tx.ino` for Transmitter) for programming the nodes.
   - Verify and upload the sketches to the respective devices. Look for the "Done uploading" message in the IDE.
3. **Operation**
   - Start the Transmitter Node and open the Serial Monitor to observe the output.
   - On the Receiver Node, open the Serial Monitor and set to 9600 baud to see the incoming data (as shown below).





Lab Assignment

1. **OLED Display Integration**
   - Display the following information on the OLED for both LoRa Receiver & Sender:
     - "Setup Successful"
     - "Setup Failed"
     - "Sending Message"
     - "Waiting for Reply"
     - “Message Received”
   - Refer to `ssd1306_i2c.ino` for OLED display implementation. Install the “Adafruit SSD1306” library first.
   - Alternatively, may refer to `ssd1306_ascii.ino`. But it uses "SSD1306Ascii" library.
2. **Enhance Peer-to-Peer Reliability**
   - Implement features to prevent crosstalk traffic interference, e.g., ACK and retransmit.
3. **Improve Message Protocol**
   - Develop ID-based messaging with a header, payload, and checksum supporting at least three devices, i.e. simple message filtering, only accepting messages that are directed to the node's agreed ID.

References

1. SemTech's LoRa Technology Overview
2. Building a LoRa-based Device with Arduino
3. CH341 Serial Driver